

ZACHARY T. MINEIKO

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EDUCATION

University of Iowa

Bachelor of Science in Engineering - Electrical Engineering

May 2025

Focus Area: Computer Science

Related Coursework: Generative AI, Communication Networks, Computer Architecture, Software Design, DSA

TECHNICAL SKILLS

Circuit Design & Simulation: Modelsim, Autodesk Fusion 360, KiCad, Micro-Cap SPICE

Development Tools: OpenCV, AVR Assembly, SolidWorks, Atmel Studio, Matplotlib, Arduino IDE, Platform IO

Languages: Python, C/C++, Embedded C, JavaScript (Node.js), C#, Verilog, Bash

WORK EXPERIENCE

FlipWyz - Full Stack Engineer (Contract)

Aug 2025 – Current

- Architect and design a production ready cross platform marketplace web app with React Native and Supabase
- Optimize PostgreSQL + PostGIS and queries to improve geospatial latency across large datasets
- Integrate Stripe for multi-party split payments, transactional integrity, and real time checkout synchronization
- Design and implement structured data flows and API integrations for listings, payments, address validation, and logistics, ensuring consistency and referential integrity across services
- Apply analytical and data driven problem solving to enhance query performance and reduce latency
- Implement modular and efficient component abstractions to reduce render overhead and improve responsiveness

Outsider - Full Stack Developer

Dec 2024 – June 2025

- Spearheaded the development of a mobile social media app using React Native, Expo, Tailwind CSS, Supabase; built core features resulting in a functional MVP within 6 weeks
- Took initiative in leading a 3-person development team, establishing sprint based workflows and managing task delegation via weekly planning meetings, which improved development efficiency by 40%
- Established standardized development workflows to improve team consistency and code quality resulting in fewer integration conflicts and a smoother sprint cycle

Firefly - Engineering Technician Intern

Chicago, IL

Mar 2022 – Aug 2022

- Gained expertise in firmware and hardware troubleshooting for digital billboards on taxis and rideshare cars
- Transformed Chicago branch performance from worst to best performing through systematic repair methodologies
- Automated data collection processes using Python and OpenCV for gaining key insights on the taxi market
- Engineered repair workflows resulting in 20% increase in daily fixed screens stressing diligence and organization
- Participated in Agile processes by contributing to weekly stand-ups and observing development sprints

Custom Craft Builders - Electrician's Assistant/Web Developer

Chicago, IL

May 2020 – May 2023

- Lead installation teams for low voltage cable systems including coaxial and Ethernet network infrastructure
- Developed, deployed, and maintained company website using HTML, CSS, JavaScript, and Bootstrap 5 framework
- Configured and managed AWS EC2 instance hosting company website, ensuring high availability and performance

PROJECT HIGHLIGHTS

Smart Pet Feeder - Senior Design Project

Dec 2024 – May 2025

- An IoT smart food tracking system for multi-pet homes to monitor and improve pets' health based on various metrics
- Developed multi-voltage ESP32 based control system with real time data processing and wireless connectivity
- Created custom PCB schematics and component footprints using Autodesk Fusion 360 for multi-sensor integration
- Integrated RFID identification system for individual pet recognition and personalized feeding schedules

IR Receiver with Hybrid Signal Filtering

Oct 2024

- Engineered and implemented a hybrid analog/digital IR receiver detecting IR interruptions using 940nm photodiode
- Designed multi-stage signal conditioning with non-inverting amplification, buffering, and a 2nd-order bandpass filter centered at ~421Hz to match existing emitter frequency constraint
- Implemented a 6th-order Chebyshev digital filter to extract signal envelope and reject ambient noise
- Validated system performance in high noise environments and achieved consistent detection up to 42 inches

Mastermind - Poker Assistant

April 2025

- Developed poker assistant using smart glasses and YOLOv8 for object detection of playing cards during gameplay
- Designed and implemented backend data flow across Raspberry Pi, FastAPI inference server (GCP), and Firebase
- Demonstrated expertise in embedded vision, AI inference, and mobile app development within an IoT system